

Django YouTube Clone

A **YouTube-style video sharing platform** built with **Django** and **ImageKit** CDN for video/thumbnail storage and streaming.

· Features

- **·Video Upload** · Upload videos with automatic CDN storage via ImageKit
 - **·Thumbnails** · Auto-generated thumbnail support
 - **·Like/Dislike** · Voting system for videos
 - **·Streaming** · Video streaming via CDN URLs
 - **·User Authentication** · Django auth system
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.. Architecture

```
Browser · Django Views · Video/VideoLike Models · SQLite DB
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ImageKit SDK ... ImageKit CDN (video + thumbnail storage)
```

· File Structure

```
Django-YouTube-Clone/
... youtube/
·   ... manage.py           # Django management
·   ... youtube/           # Project settings
·   .   ... settings.py
·   .   ... urls.py
·   ... videos/
·       ... models.py       # Video, VideoLike models
·       ... views.py        # CRUD, upload, like/dislike views
·       ... urls.py         # URL routing
·       ... templates/      # HTML templates
... pyproject.toml
... uv.lock
```

· Getting Started

Prerequisites

- Python 3.10+, ImageKit account

```
cd Django-YouTube-Clone
uv sync
cd youtube
python manage.py migrate
python manage.py runserver
```

Environment Variables (.env)

```
IMAGEKIT_PRIVATE_KEY=...  
IMAGEKIT_PUBLIC_KEY=...  
IMAGEKIT_URL_ENDPOINT=...
```

· Logic Flow

- 1 **Upload** · User uploads video file via form
- 2 **CDN Storage** · Video uploaded to ImageKit CDN, URL stored in DB
- 3 **Model Creation** · Video model created with title, description, CDN URLs
- 4 **Streaming** · Video served via CDN streaming URL
- 5 **Interaction** · Users can like/dislike (one vote per user per video)
- 6 **Display** · Video feed with thumbnails and metadata

Key Models

Model	Fields	Purpose
Video	title, description, video_url, thumbnail_url, created_at	Video content
VideoLike	video (FK), user (FK), is_like	Like/dislike votes

· Dependencies

- Django >= 6.0
 - imagekitio
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· License

Educational project · use freely for learning and reference.